

A novel abnormal pore pressure prediction framework based on machine learning and drilling-rock mechanics parameter data

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ABSTRACT

The Xihu Sag in the East China Sea Basin exhibits complex geology (e.g., deep faulting and fragmented pressure systems) and anomalous overpressure with pressure reversal in target formations, posing severe challenges to conventional pore pressure prediction methods. This study proposes an integrated framework for pore pressure prediction that fuses nine drilling parameters with three rock mechanics parameters. The framework overcomes the common limitation of previous studies relying on a single data source (well log or seismic data). Model development and validation are demonstrated using 8 vertical wells across three blocks in the Xihu Sag. Five Machine Learning (ML) algorithms (Back Propagation, K-Nearest Neighbor, Support Vector Regression, CatBoost, LightGBM) are optimized, with a detailed investigation of dataset partitioning strategies (direct vs. randomized division) to eliminate high-pressure prediction bias. Comprehensive evaluation via six metrics (Mean Squared Error (MSE), Mean Absolute Error (MAE), etc., including training time) demonstrates that the LightGBM model outperforms others: it achieves near-perfect fitting ($R^2=0.999$), minimal prediction errors ($MSE=0.001$, $MAPE=0.239\%$), and short training time (0.45 s), with a narrow relative error range (-1.83% to $+1.61\%$) for both normal and overpressure zones. Practical validation on an adjacent well and four regional wells confirms its robust generalization (average accuracy $>97\%$), with its prediction results consistent with the "normal pressure-overpressure-pressure reversal" distribution pattern of the target block. Importantly, the ML model is successfully applied to two adjacent structural units and, without any parameter adjustment, maintained prediction accuracy above 92%. This work provides a high-precision, field-applicable pore-pressure prediction tool that mitigates drilling risks and offers a reproducible reference framework for pore pressure prediction in other similar complex overpressured basins.

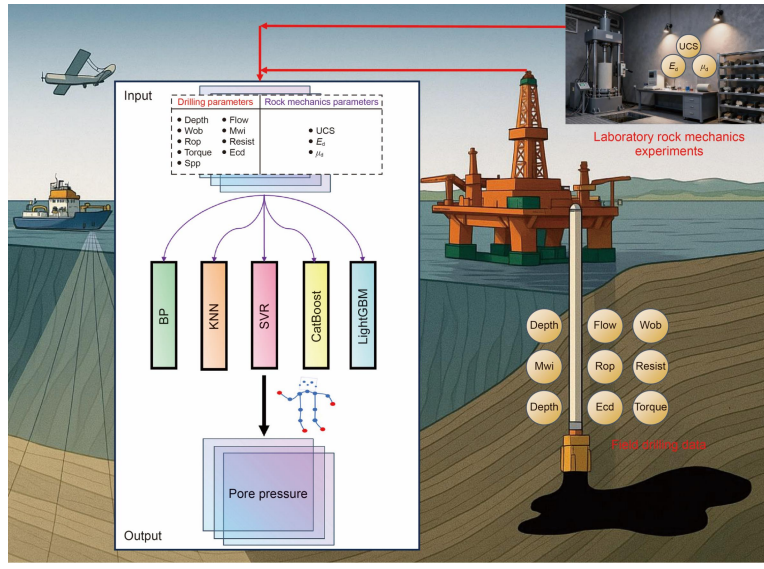
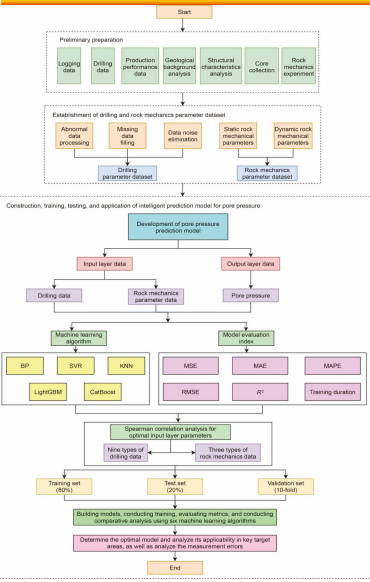


Fig. 1. Technology roadmap.

Fig. 2. Schematic diagram of model construction process.

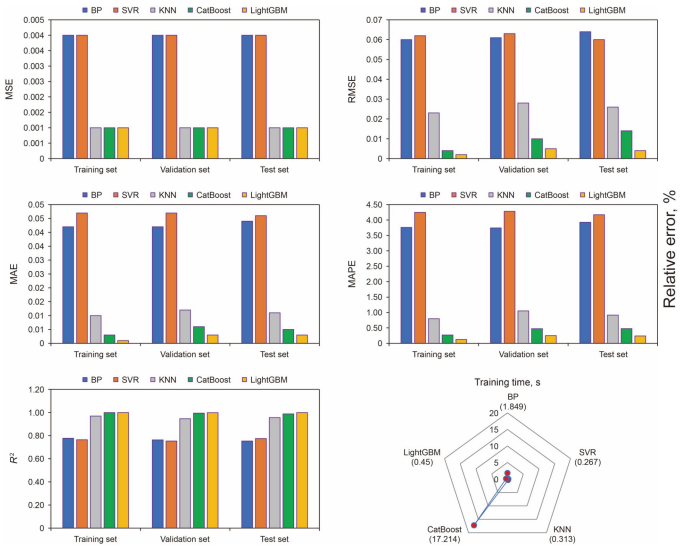


Fig. 3. Evaluation results of the 5 algorithm models.

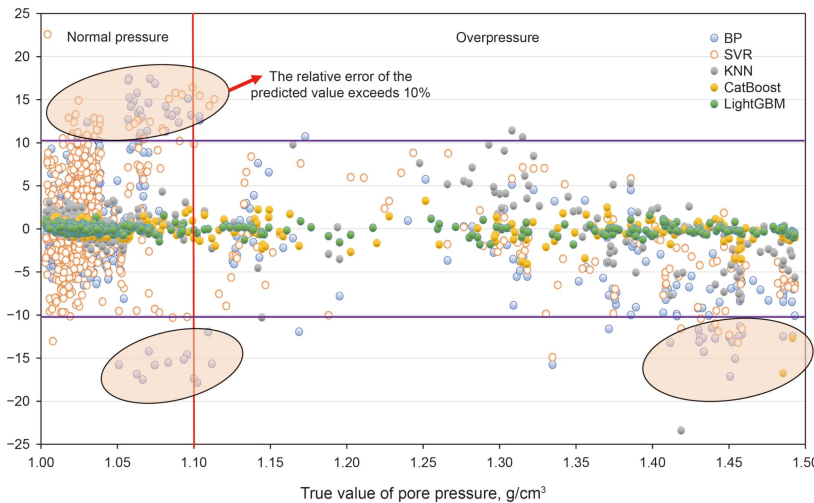


Fig. 4. Performance of the five algorithm models on the test data.

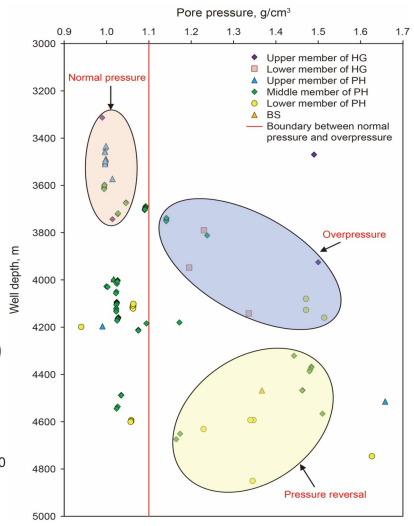


Fig. 5. Vertical distribution map of measured pore pressure data.